TRAMADOL

Seizure risk

Controlled in many countries

CHEMICAL NAME (1R,2R)-2-[(dimethylamino)methyl]-1-(3-methoxyphenyl)cyclohexan-1-ol (as hydrochloride, racemic)

CLASS Weak (atypical) opioid analgesic with SNRI activity

TRADE NAMES Tramal · Ultram · Contramal · Tramadex

THERAPEUTIC USES / INDICATIONS

- Moderate to severe pain: post-operative, musculoskeletal, neuropathic.
- Chronic pain when NSAIDs are unsuitable.
- Combined with paracetamol for additive analgesia.

PHARMACODYNAMICS

- Weak **μ-opioid** agonist (mainly via metabolite).
- Inhibits **serotonin and noradrenaline reuptake** → enhances descending pain inhibition.
- Less respiratory depression and constipation than morphine.

ADMINISTRATION

- **PO:** capsules, tablets (immediate- or extended-release; swallow ER tablets whole).
- **IV:** slow injection over 2–3 min or infusion; **IM**.
- May be given with or without food.

DOSAGES

- **Adult:** 50–100 mg every 4–6 h; max 400 mg/day.
- **Elderly >75 yr:** max 300 mg/day.
- **CrCl <30 mL/min:** every 12 h; max 200 mg/day.
- **Children ≥12 yr:** 1–2 mg/kg per dose (avoid <12 yr and post-tonsillectomy).
- **ER:** 100–200 mg every 12 h.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Peak:** 2 h. **Duration:** 6 h.
- Metabolised by CYP2D6 to O-desmethyltramadol (more potent opioid); ultra-rapid metabolisers at risk of toxicity.
- Excreted in urine. **t½:** 6 h (7.5 h for active metabolite).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, dizziness.
- Constipation, dry mouth, sweating.
- Drowsiness, headache.

ADVERSE EFFECTS (SERIOUS)

- Seizures (dose-related; lower threshold).
- Serotonin syndrome with serotonergic drugs.
- Respiratory depression (high doses, CYP2D6 ultra-rapid metabolisers).
- Hypoglycaemia, hyponatraemia; dependence and withdrawal.

CONTRAINDICATIONS & CAUTIONS

- Epilepsy not controlled by treatment.
- Use with MAOIs or within 14 days.
- Acute intoxication with alcohol, hypnotics or other opioids.
- Children <12 yr; breastfeeding (CYP2D6 risk).

KEY INTERACTIONS

- **SSRIs, SNRIs, TCAs, MAOIs, linezolid, triptans:** serotonin syndrome; ↑ seizure risk.
- **Carbamazepine:** ↓ tramadol effect.
- **Warfarin:** ↑ INR.
- **CNS depressants:** additive sedation.

NURSING MANAGEMENT

- Ask about epilepsy and antidepressant use.
- Monitor pain score, RR, sedation and BP.
- Give antiemetic if nausea; IV slowly to reduce nausea.
- Watch for agitation, sweating, tremor, fever (serotonin syndrome).
- Assess for misuse or dependence.

PATIENT EDUCATION

- Do not exceed the prescribed dose; risk of fits.
- Avoid alcohol; may cause drowsiness.
- Tell the doctor about any antidepressants.
- Do not stop suddenly after long use.

KEY POINTS

- Dual action: weak μ-agonist + SNRI.
- Risks: seizures and serotonin syndrome.
- Max 400 mg/day; lower in elderly and renal failure.
- CYP2D6 metabolism — variable response.

SOURCES CHECKED **[38]** Tramadol 50 mg/ml solution for injection/infusion (UK SmPC) **[39]** Tramadol 100 mg/ml oral drops, solution (UK SmPC) **[40]** Tramadol Hydrochloride, oral (US label) **[41]** Tramadol Hydrochloride, oral (US label, Proficient Rx LP)

Full references p. 432–457